

$\pi_{0.5}$ · arXiv 2504.16054 · Physical Intelligence

Stakeholder

18+3 · seat 2

Open-world KPI = new homes

OpenPI $\pi_{05_}$ * · BEHAVIOR'26 baseline

Sep 30 paper club · VLA Architectures

Spine: open-world + hybrid FAST discrete pretrain → FM retained

Semantic subtask → FM chunk (~50-step/~1s @ 50 Hz) · NOT a classical planner · RTC companion

Club spine · one-liner

Open-world VLA via heterogeneous co-training + hierarchical inference:

FAST discrete pretrain → FM retained for continuous control.

Semantic subtask → ~50-step / ~1s FM chunk @ 50 Hz. NOT a classical planner.

RTC companion for async soft-inpaint. Blogs: “FAST replaced FM” = FALSE.

BEHAVIOR'26: $\pi 0.5$ + GR00T baselines · Oct 2 = $\pi 0.5$ -baseline + deltas.

Open-world KPI · new homes, not one kitchen

OLD KPI

Skill in one matched kitchen

lab demos · train≈test scenes

$\pi_{0.5}$ KPI

Generalization across new homes

~400h MM · ~100 homes · co-train leverage

OpenPI

pi05_base
pi05_libero
pi05_droid

BEHAVIOR'26

official baseline
+ GROOT N1.7
submit 10/16/2026

'25 proof

RLC winner
 $q \approx 0.260$ on $\pi_{0.5}$
validates substrate

Open-world KPI · new homes, not one kitchen

OLD KPI

Skill in one matched kitchen

lab demos · train≈test scenes

$\pi_{0.5}$ KPI

Generalization across new homes

~400h MM · ~100 homes · co-train leverage

OpenPI

pi05_base
pi05_libero
pi05_droid

BEHAVIOR'26

official baseline
+ GROOT N1.7
submit 10/16/2026

'25 proof

RLC winner
 $q \approx 0.260$ on $\pi_{0.5}$
validates substrate

FM vs FAST · say this aloud

Hybrid FAST discrete pretrain → FM retained for continuous control

Claim	Verdict
FAST replaced FM	MYTH
FAST pretrain + FM post-train	TRUE
FM head retained at inference	TRUE
OpenPI FM API for $\pi_{0.5}$	CLEAR
FAST is the deployed LL controller	FALSE

OpenPI ships FM head for $\pi_{0.5}$ · KI (2505.23705) formalizes stop-grad FAST+FM

Budget decision · what to fund

Fund

FM fine-tunes on public pi05_*

OpenPI / Comet / BEHAVIOR stacks

Fund

Optional RTC for latency loops

Companion to FM policies

Fund

HL / stage tracking + VI labels

Paper + '25 System-2 moral

Ignore

“FAST replaced FM” roadmaps

Hybrid; FM retained at deploy

Defer

Private 280k+80k MM homes recipe

Irreproducible at club scale

Co-training mixture · MM ME CE HL WD VI

97.6% of phase-1 examples are NOT target mobile household teleop

MM

~400h / ~100 homes

Target mobile manip

ME

Static multi-env

Cross-emb + diversity

CE

Lab + OXE

Task diversity

HL

Subtask (+boxes)

Hierarchy supervision

WD

Web VQA/caption

OOD semantics

VI

~11% of HL-MM

Verbal coaching

Hybrid recipe · FAST discrete pretrain → FM retained

Bust the myth: FAST does NOT replace FM at deploy

① PRE-TRAIN · 280k steps

Discrete next-token VLA

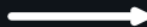
SigLIP 400M + Gemma ~2B

text · boxes · FAST action tokens · $\alpha=0$

FAST action tokens

scalable discrete pretrain signal

Mixture: MM · ME · CE · HL · WD



② POST-TRAIN / INFER · 80k

Joint NTP + flow matching $\alpha=10$

Action expert ~300M · FM

Ada-RMSNorm on τ · random init expert

Inference: AR subtask → 10 FM steps

~50-step / ~1s chunk @ 50 Hz PD

MM+ME+WD+VI · drop CE · FM RETAINED

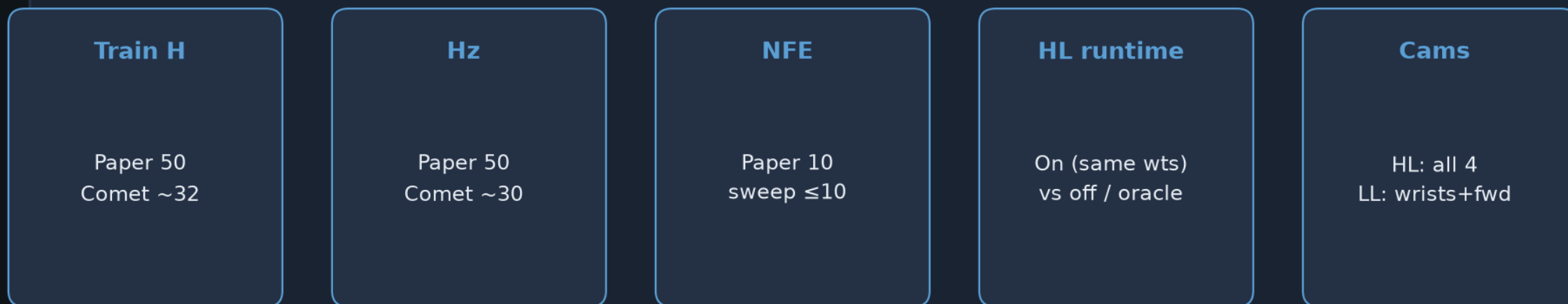
NOT classical planner · same model for HL + LL · RTC = async soft-inpaint companion (2506.07339)

Hierarchical control loop · semantic subtask → FM chunk

Outer semantic loop · inner continuous control · NOT a classical planner



Control knobs (paper vs challenge priors)



2026 BEHAVIOR map · CLEAR

Official baselines · scale · timeline · '25 preview

Official baselines

π 0.5 + GROOT N1.7

behavior.stanford.edu/challenge

Scale

100 tasks · 7 scenes · 20k demos

~1950 h · RGB+depth+proprio · BDDL

Timeline 2026

07/02



launch

10/16



submit

11/04



winners

'25 winner on π 0.5 · preview of '26 bottlenecks

RLC adapted π 0.5 → ~26% q (correlated FM noise · System-2 stages · multi-sample FM · compression)

Bottlenecks: precision grasps · non-Markov stages · recovery demos · Oct 2 → propose deltas on this substrate

Oct 2 proposal framing · π 0.5-baseline + targeted deltas

NOT greenfield · mirrors '25 winning pattern + challenge starter kit

1

Keep FM

FAST/KI for reps & language
never “replace FM with FAST”

HIGH
LEVERAGE

2

Hierarchy / stages

Paper HL + '25 System-2
non-Markov BEHAVIOR states

HIGH
LEVERAGE

3

RTC

Async soft-inpaint for delayed
control loops (2506.07339) address demo manifold narrowness

HIGH
LEVERAGE

4

Recovery / VI

Intervention + verbal coaching
address demo manifold narrowness

HIGH
LEVERAGE

5

Cross-emb + co-train

Web/HL when scene diversity
limited vs real homes

HIGH
LEVERAGE

Apps: new-home cleaning · language tidying · coachable robots · OpenPI FT on DROID/LIBERO/custom

$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization

Physical Intelligence

Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, Ury Zhilinsky

<https://pi.website/blog/pi05>



Fig.5 · Robot system 18-19 DoF · 4 cams

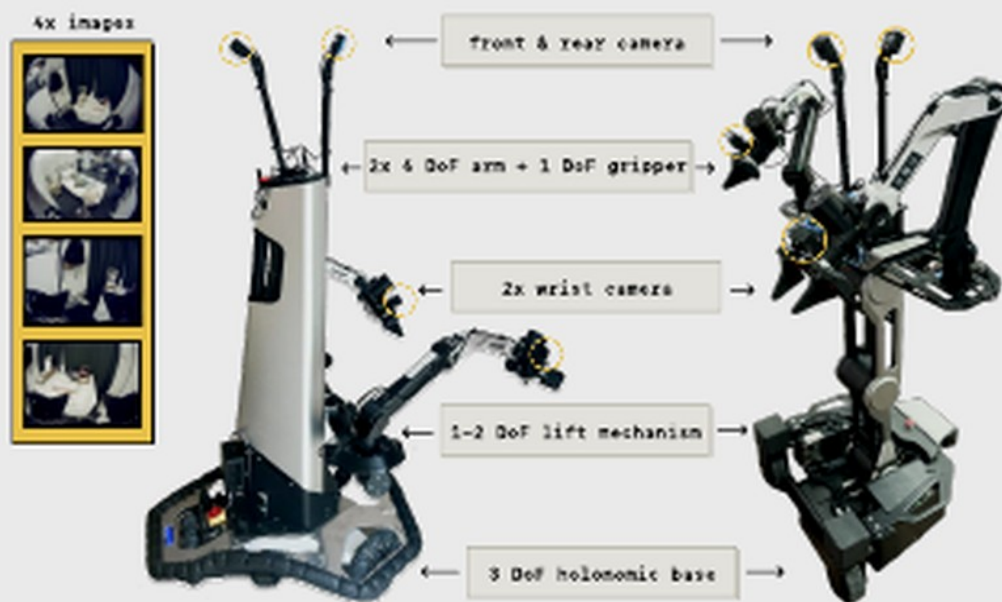


Fig. 5: **Robot system overview.** We use two mobile manipulator platforms – each has four cameras (forward, backward, and both wrists), two 6 DoF arms with parallel jaw grippers, a mobile base, and a torso lift mechanism. The $\pi_{0.5}$ model controls the joints and grippers of each arm, base velocity, and the lift position, resulting in 18-19 DoF state and action spaces.

The control system is very simple: the $\pi_{0.5}$ model directly commands target poses for the arms, gripper, and torso lift, and the target base velocities at 50 Hz (with action chunking). These targets are tracked with simple PD controllers, without any additional trajectory planning or collision detection. All

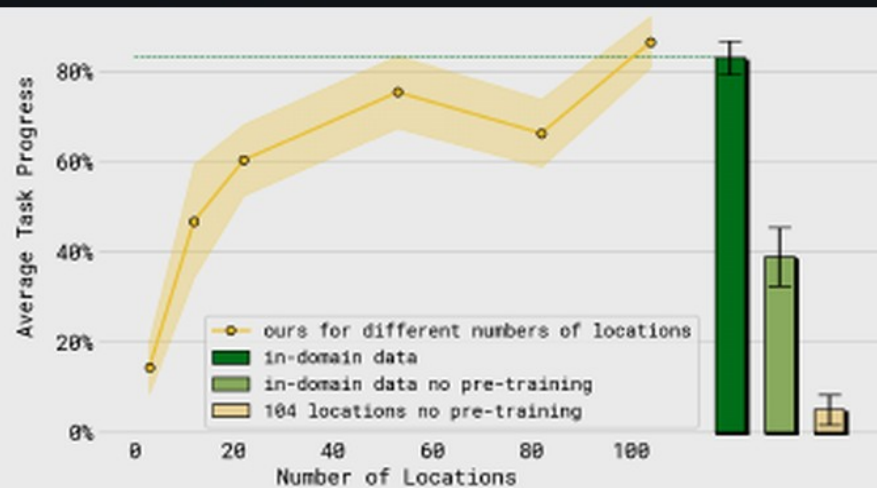


Fig. 8: **Evaluating performance with different numbers of locations.** Performance over the four test tasks — “dishes in sink”, “items in drawer”, “laundry basket”, “make bed” — improves with more training environments. The dashed green line and green bar show a baseline model that includes the test homes in the training set. Compared to this model, our best model achieves similar performance, despite not seeing any data from the test homes.

those in the training data (but new instances), as well as out-of-distribution objects from unseen categories. The latter necessitates broad semantic generalization.

The results of the first experiment are shown in Figure 8. The average performance among the tasks generally improves with more training locations. To quantify how much the final model (with 104 locations) bridges the generalization gap, we include a control (shown in green) that is trained directly on data from the test homes. This control attains similar performance as the final 104-location model, suggesting that our co-training recipe effectively enables broad generalization, reaching similar performance to a model trained on the test

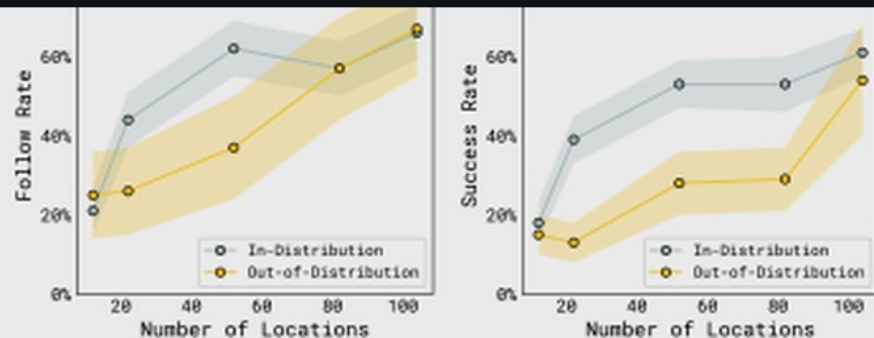


Fig. 9: **Evaluating language following with different numbers of training locations.** We evaluate language following rate and success rate for picking up user-indicated items and placing them into drawers or sinks, averaged over seen object categories (“in-distribution”) or unseen categories (“out-of-distribution”). Performance increases steadily as we increase the number of training locations.

of locations in the training data increases, both language following performance and success rate improve. As expected, the performance on in-distribution objects improves more quickly than that of out-of-distribution objects. As each new environment introduces new household items, the model becomes generally more robust and starts to generalize to task categories that were not present in the training data.

C. How important is each part of our co-training recipe?

To study Question (3), we compare our full $\pi_{0.5}$ model to other training mixtures to study the importance of each mixture component, again using end-to-end task performance in the mock homes and the language following evaluation described in Section V-B. As a reminder, our full recipe uses data from mobile manipulators in many environments (MM), static manipulators in many environments (ME), and diverse

Apps + caution · Visionary north star

Applications

- ▶ New-home mobile cleaning
- ▶ Language-conditioned tidying
- ▶ Coachable robots (VI)
- ▶ OpenPI FT: DROID / LIBERO / custom
- ▶ Richer prompts / longer memory (paper Discussion)

Caution

- ▶ Real-home success $\neq$ solved BEHAVIOR
- ▶ BDDL goals · depth · 100-task multitask
- ▶ Hard handles · occlusion · HL loops
- ▶ Private MM recipe unreproducible
- ▶ Still clearest shared substrate for '26

One-slide message

Open-world household VLA is a co-training + hierarchy problem,
not pure scaling. 2025 operationalized the backbone; 2026 should stress-test
held-out semantics, memory, and teachable HL — GPU-honest, not private-home cosplay.

π_0 VS $\pi_{0.5}$ · what changed

π_0 (2410.24164)

- FM action expert throughout
- No FAST pre-phase
- No unified HL head in recipe
- Lab / matched scenes focus
- ~10k h cross-embodiment
- H=50 · ~10 Euler · 20-50 Hz
- PaliGemma + ~300M expert

$\pi_{0.5}$ (2504.16054)

- FAST pretrain → FM post-train
- FM RETAINED at inference
- Unified HL subtask → LL chunk
- Open-world new homes KPI
- ~400h MM / ~100 homes + mix
- H=50 @ 50 Hz · 10 FM steps
- Co-train: MM ME CE HL WD VI

Novelty = mixture curriculum + hierarchy + hybrid recipe + home-scale eval — not new FM math

Stakeholder · close

KPI = new homes. OpenPI pi05_* + BEHAVIOR'26 official baseline.

'25 winner on Pi0.5 validates substrate · ignore FAST-replaced-FM myths

Q&A · Sep 30 · $\pi_{0.5}$ / 2504.16054

arXiv:2504.16054v1 [cs.LG] 22 Apr 2025

$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization

Physical Intelligence

Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolò Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haobuan Wang, Lili Yu, Ury Zhilinsky
<https://pi.website/blog/pi05>

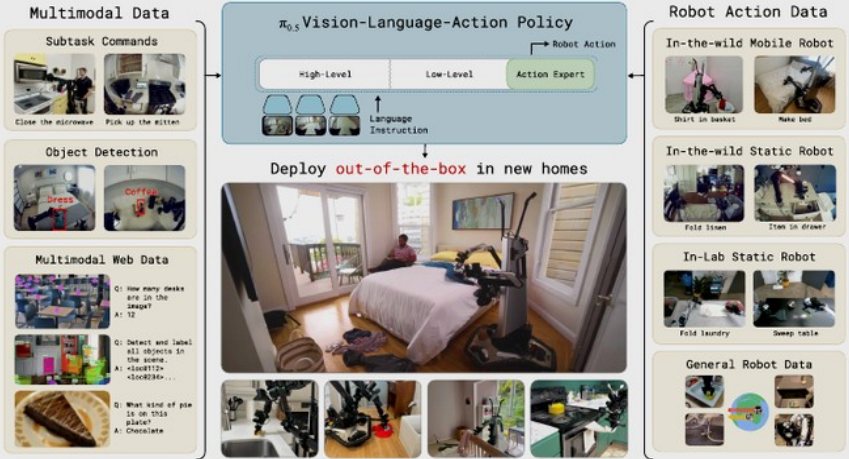


Fig. 1: The $\pi_{0.5}$ model transfers knowledge from a heterogeneous range of data sources, including other robots, high-level subtask prediction, verbal instructions, and data from the web, in order to enable broad generalization across environments and objects. $\pi_{0.5}$ can control a mobile manipulator to clean kitchens and bedrooms in new homes that were not present in the training data, performing complex multi-stage behaviors with durations of 10 to 15 minutes.

Abstract—In order for robots to be useful, they must perform practically relevant tasks in the real world, outside of the lab. While vision-language-action (VLA) models have demonstrated impressive results for end-to-end robot control, it remains an open question how far such models can generalize in the wild. We describe $\pi_{0.5}$, a new model based on π_0 that uses co-training on heterogeneous tasks to enable broad generalization. $\pi_{0.5}$ uses data from multiple robots, high-level semantic prediction, web data, and other sources to enable broadly generalizable real-world robotic manipulation. Our system uses a combination of co-training and hybrid multi-modal examples that combine image observations, language commands, object detections, semantic subtask prediction, and low-level actions. Our experiments show that this kind of knowledge transfer is essential for effective generalization, and we demonstrate for the first time that an end-to-end learning-enabled robotic system can perform long-

horizon and dexterous manipulation skills, such as cleaning a kitchen or bedroom, in entirely new homes.

I. INTRODUCTION

Stuff your eyes with wonder... See the world. It's more fantastic than any dream made or paid for in factories.

Ray Bradbury, *Fahrenheit 451*

Open-world generalization represents one of the biggest open problems in physical intelligence: embodied systems such as robotic arms, humanoids, and autonomous vehicles only truly become useful when they can leave the lab and handle the diverse situations and unexpected events that occur in the real world. Learning-based systems offer a path to en-